SMART KNOB DISPLAY MODULE SPECIFICATION

2.1 Inch Smart Knob Display with TOUCH	
Product:	SFG-VIEWE-DesKnob
MCU：	ESP32-S31NRV16
PCB：	Knob Module Core Board S31v1 + Dock Board S31v1
Display：	2.1" round 360×360, GC9B72NA + CST826, QSPI
Version:	V1.0
Date:	2026-9-17

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Customer Confirmation

Approved by	Notes

REVISION HISTORY

Revision	Date	Contents of Revision Change	Remark
V1.0	20260917	Initial release	

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1. Introduction



Figure 1-1: Product Appearance (The above are sample pictures; the actual product shall prevail, and there may be differences in shell color)

SFG-VIEWE-DesKnob is a high-performance AI smart knob display module in the VIEWE housed finished-goods series (SFG prefix). It is based on the Espressif ESP32-S31NRV16 SoC (QFN80, 16 MB in-package 1.8 V Octal PSRAM; on-board external W25Q128 Flash) and a 2.1-inch round capacitive touch panel (resolution 360×360, display driver IC GC9B72NA, touch IC CST826, host-side QSPI interface). Hardware uses a dual-PCB architecture (Knob Module Core Board S31v1 + Knob Module Dock Board S31v1) linked by board-to-board FPC2. Core board + dock board as a complete module. The product integrates rotation detection (Alps SSCM two-way detector switches, net names EC1/EC2), a 12-LED WS2812-compatible ring, ES8311 audio codec + NS4150B amplifier + analog microphone, Type-C power/download, TP4055 Li-ion charging, and MicroSD expansion (schematic includes DRV2605L haptics; currently not fitted / not used).Customers can fabricate base plates according to their own requirements to unlock more possibilities for the product. Typical uses include smart-home central knobs, voice-assistant front ends, and lightweight HMI.

Notice: ESP32-S31 is currently a preview device. ESP-IDF builds must use the --preview option (for example: idf.py --preview set-target esp32s31). Module electrical and RF ratings follow the Espressif ESP32-S31 series module datasheet (pre-release). New designs that need Wi-Fi 6, native USB, and on-board codec audio should use this S31 knob solution.

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1.1 Product Features

CPU / Memory / Wireless / Security:

1.Processor (ESP32-S31NRV16)

1) SoC architecture: high-performance HP system + low-power LP system.
2) HP: 32-bit dual-core RISC-V, up to 320 MHz; single-precision FPU on each core; SIMD on one core; dual-core CoreMark® @320 MHz: 2195.20 (6.86/MHz).
3) LP: 32-bit single-core RISC-V, up to 40 MHz, 2-stage pipeline; remains alive in Deep-sleep.
4) Package: QFN80 (8×8 mm); chip ambient −40～85 °C (N = normal temperature grade).
5) Board antenna: schematic includes on-board antenna (CA-C03) and optional IPEX external antenna connector.

2.Memory

1) On-chip: 320 KB ROM, 512 KB HP SRAM, 32 KB LP SRAM.
2) In-package PSRAM: 16 MB, 1.8 V Octal SPI (encapsulated; pins not brought out; R=in-package PSRAM, V=1.8 V, 16=16 MB). Do not confuse with ESP32-S31NRV32 (32 MB)—this product uses NRV16.
3) On-board external Flash: W25Q128JVPIQ (128 Mbit = 16 MB). Schematic note: R4 10K to GND means 1.8 V flash; VDD_PSRAM_1P8_* power domain present → Flash/PSRAM supplied in the 1.8 V domain. W25Q128 is external Flash, not PSRAM.
4) Parallel access to flash and PSRAM supported; external flash SPI/Dual/Quad/QPI (chip capability up to 256 MB).

3.Wireless

1) 2.4 GHz Wi-Fi 6 (IEEE 802.11ax, 20 MHz-only non-AP; also b/g/n), 1T1R.
2) Bluetooth® 5.4 (LE) + Bluetooth® Classic.
3) Zigbee 3.0 and Thread 1.4 (IEEE 802.15.4).

4.Security (chip-level summary)

1) Secure Boot; off-chip memory encryption/decryption (XTS-AES for Flash/PSRAM).
2) Crypto accelerators: AES / SHA / RSA / ECC / HMAC, digital-signature peripherals, TRNG, key manager.
3) APM and TEE, among others (see official datasheet).

5. Peripheral interfaces ([Dock] + [Core])

1) [Core] MicroSD (TF-018), 4-bit SDMMC
2) [Core] QSPI Display
3) [Core] Rotation detection: 2 bidirectional detector switches (not EC11 incremental encoders); EC1 connects to GPIO2/3, EC2 via FPC2

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4) [Core + Dock] 33-pin board-to-board FPC connector: interconnection between core board and dock; exposed as expansion interface for standalone core board，available for customer expansion
5) [Dock] USB-C: 5V power input, firmware download and debug; USB 2.0 High-Speed OTG + USB Serial/JTAG, no CH340
6) [Dock] TP4055 single-cell Li-ion charging, 2.2 kΩ PROG resistor, charge current ≈ 455 mA
7) [Dock] Side slide power switch + BOOT download key
8) [Dock] 12 RGB LED ring
9) [Dock] ES8311 audio codec + NS4150B amplifier + ZTS6216 analog MIC + speaker connector CN2

Display

1) Size: 2.1 Inch (round)
2) Resolution: 360 × 360
3) Display module P/N: ALL-UE021HV-RU39-A021A
4) Interface: QSPI (as wired on the core board; not parallel RGB)
5) Driver IC: GC9B72NA
6) Touch IC: CST826 (multi-touch capacitive, G+F CTP)
7) Brightness: 300 (min) / 350 (typ) cd/m^2
8) Logic supply: 3.3 V
9) Optical/mechanical display details are in Section 2.3.

Other

10) Operation Temperature: −20 ~ 70 °C (limited by the LCD).
11) Storage Temperature: −30 ~ 80 °C.
12) MCU module rating: −40 ~ 85 °C.

1.2 Applications

With wireless connectivity, a round touch display, rotary interaction, and local audio/haptic capability, VIEWE-DesKnob is suitable for:

- Smart-home central control knobs / scene panels
- Controllers for speakers, lighting, HVAC, and other appliances
- Conference-table / desktop office controllers
- Industrial lightweight HMI / parameter knobs
- AI voice-assistant front ends (local MIC + speaker)

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- Maker, education, and prototyping platforms

1.3 Product Naming

Official product name: SFG-VIEWE-DesKnob

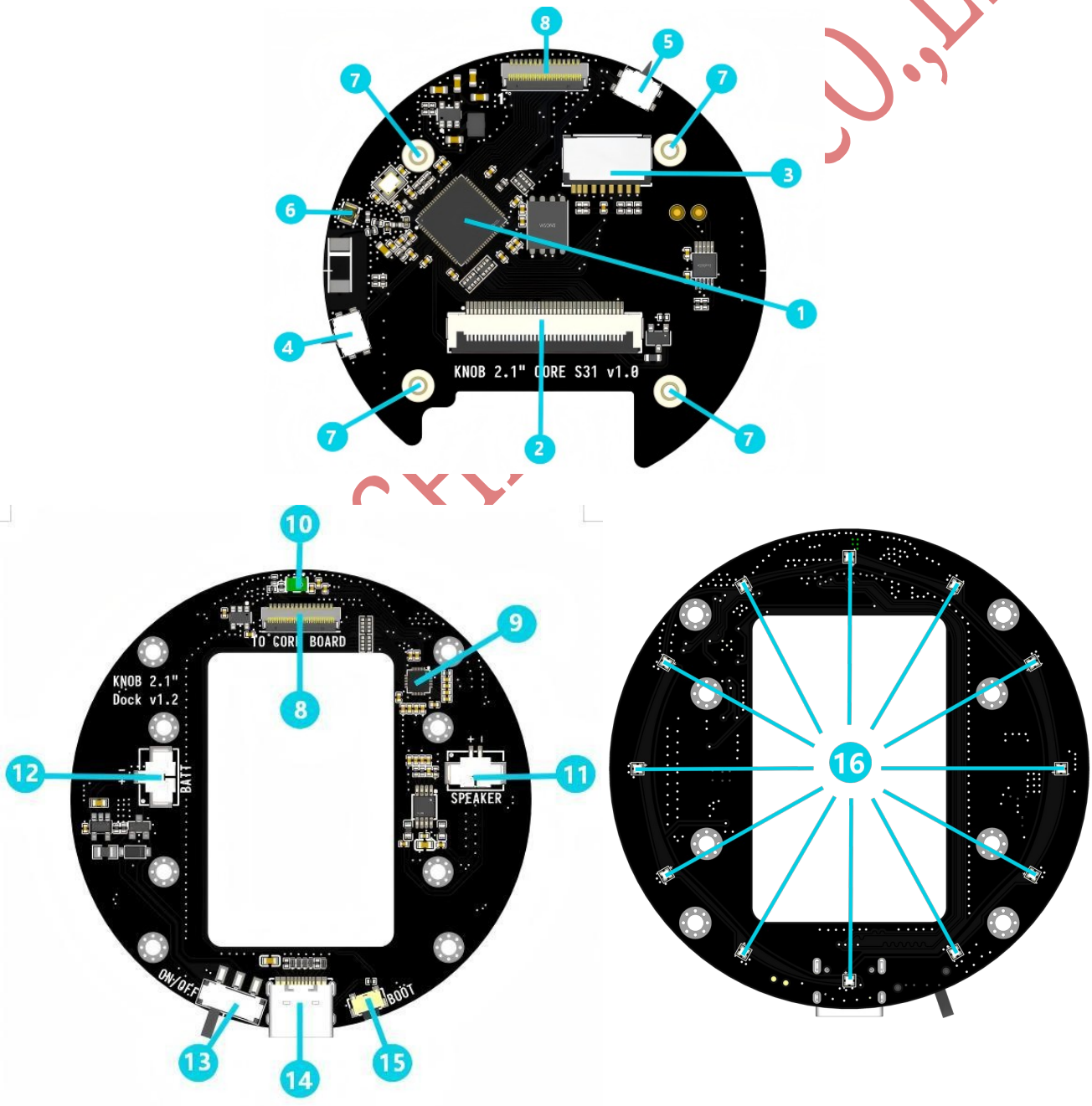
Field	Code	Meaning
Form	SFG	Semi-Finished Goods
Brand	VIEWE	VIEWE brand
Structural features	DesKnob	Desk+Knob, desktop knob device

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2.Product information

2.1 Interface Description

This product is a cylindrical smart knob: a 2.1" round touch glass on top, a translucent LED ring in the middle, and Type-C, a slide power switch, microphone opening, and speaker opening on the side. Internally it stacks as Rear housing → Li-ion Battery & Speaker → Dock Board → Light Guide Ring → Support Bracket → Rotating Bearing → Metal Housing → Core Board → 2.1" Round Touch Screen. (see §5).



Appearance vs. interface mapping:

1. Main control:

ESP32-S31NRV16 (HP dual-core RISC-V up to 320 MHz + LP single-core), 16 MB in-package PSRAM + on-board external W25Q128 (16 MB) Flash, on-board/IPEX antenna.

2. Display interface & Touch interface(FPC1):

40-pin (FPC-05F-40PH20), QSPI LCD + I2C touch (TP_SCL/SDA/RST/INT); backlight controlled by LCD_BLK via AO3400A (Q2); See §2.1.

3. SD Card Slot:

SPI interface connected for external storage expansion; See §2.8.

4. Rotation detection (two-way detector switches / net names EC1):

Core-board EC1 (GPIO2/3, 10 kΩ pull-ups); See §2.7.2.

5. Rotation detection (two-way detector switches / net names EC2):

Core-board EC2 (ESP32 not detected, connected via FPC2); See §2.5.

6. External Antenna:

The device is equipped with an antenna.

7. Positioning post

8. Board-to-board FPC2:

Core↔dock; Customers can customize dock boards according to PIN definitions to create more possibilities. ; see §2.5.

9\10\11. [Dock] Audio:

ES8311 + NS4150B + (10)MIC (ZTS6216) + (11)speaker connector CN2 (1.25-2P).See §2.4.

12. [Dock] Battery connector CN1:

1.25-2P, single-cell Li-ion; Voltage: 3.7V; Capacity: 750mAh;See §2.8.

13. [Dock] Power slide switch SW3:

MST-12D18G4, controls the VCC_OUT path; See §2.7.3.

14. [Dock] USB Type-C:

On the dock board; 5 V power and native USB download/debug (no UART bridge IC); See §2.6.

15. [Dock] BOOT key:

Dock board SW1 (GT-TC018B-H035-L1), pulls BOOTMODE (GPIO61) low to enter download mode; See §2.7.3.

16. [Dock] RGB LED ring:

12× XL-1010RGBC-2812B, data net RGB_DATA (GPIO61); See §2.7.4.

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Dual-board architecture

Board	Silkscreen / schematic name	Main functions
Core board	Knob Module Core Board S31v1	MCU, display/touch FPC, haptics, rotation detection EC1 (and EC2 breakout), MicroSD, 3V3 buck, FPC2
Dock board	Knob Module Dock Board S31v1	Type-C, charging, power switch, audio, LED ring, BOOT, FPC2

Display interface (FPC1, core-board QSPI wiring)

Connector: FPC-05F-40PH20 (40-pin). The table below lists signals already brought out on the core-board schematic; remaining pins are NC / GND / 3V3 on the display-module side, etc. Do not copy parallel DB pin definitions from the panel mechanical drawing.

Pin No.	Symbol	I/O	Description
2	TP_RST	I/O	Touch reset → GPIO8
3	TP_SCL	I/O	Touch I2C SCL → GPIO9 (shared with HAPTIC)
4	TP_SDA	I/O	Touch I2C SDA → GPIO10 (shared with HAPTIC)
5	TP_INT	I	Touch interrupt → GPIO11
16	LCD_RST	O	LCD reset → GPIO12
26	LCD_QSPI_D3	I/O	QSPI data 3 → GPIO13
27	LCD_QSPI_D2	I/O	QSPI data 2 → GPIO14
28	LCD_QSPI_D1	I/O	QSPI data 1 → GPIO15
29	LCD_QSPI_D0	I/O	QSPI data 0 → GPIO16
32	LCD_QSPI_CS	O	QSPI chip select → GPIO17
33	LCD_QSPI_SCL	O	QSPI clock → GPIO18
35	LCD_TE	I	Tearing Effect → GPIO19
38	LEDK	P	Backlight cathode (via R30=10 Ω + Q2 AO3400A to GND)
39	LEDA	P	Backlight anode (to 3V3)
Other	GND/3V3/NC	—	Per display module and core-board copper

I: input; O: output; P: power. Backlight enable LCD_BLK = GPIO7 (active high; gate has 10 kΩ pull-down).

Touch interface summary

Signal	GPIO	Description
TP_RST	GPIO8	Touch reset
TP_SCL	GPIO9	I2C SCL, 2.2 kΩ pull-up to 3V3
TP_SDA	GPIO10	I2C SDA, 2.2 kΩ pull-up to 3V3
TP_INT	GPIO11	Touch interrupt
CST826	—	Address:0x15

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address, etc.		

2.2 GPIO Definition

GPIO Number	Current Usage	Function Description
GPIO0-1	Reserved	Not assigned the IO;
GPIO2	EC1_B	Rotation detection (two-way detector switch) channel B (core board)
GPIO3	EC1_A	Rotation detection (two-way detector switch) channel A (core board)
GPIO4-6	Reserved	Not assigned the IO;
GPIO7	LCD_BLK	Backlight enable (AO3400A gate), active high
GPIO8	TP_RST	Touch reset
GPIO9	TP_SCL	I2C clock (touch)
GPIO10	TP_SDA	I2C data (touch)
GPIO11	TP_INT	Touch interrupt
GPIO12	LCD_RST	LCD reset
GPIO13-16	LCD_QSPI_D3	QSPI data 3
GPIO13-16	LCD_QSPI_D2	QSPI data 2
GPIO13-16	LCD_QSPI_D1	QSPI data 1
GPIO13-16	LCD_QSPI_D0	QSPI data 0
GPIO17	LCD_QSPI_CS	QSPI chip select
GPIO18	LCD_QSPI_SCL	QSPI clock
GPIO19	LCD_TE	LCD TE
GPIO20–23	SD-D0–D3	SDMMC 4-bit data
GPIO24	SD-CLK	SDMMC clock
GPIO25	SD-CMD	SDMMC command
USB_DP / USB_DM	USB 2.0 HS PHY	Module native USB differential, via FPC2 to Type-C
GPIO33	USB_DN	Via FPC2 to Type-C; coexist with ESP_USB_*;
GPIO34	USB_DP	Via FPC2 to Type-C; coexist with ESP_USB_*;
GPIO35	AUDIO_I2C_SCL	ES8311 I2C SCL (dock)
GPIO37	FPC2 breakout	-
GPIO38	FPC2 breakout	-
GPIO39	FPC2 breakout	-
GPIO40	AUDIO_I2C_SDA	ES8311 I2C SDA
GPIO42	I2S_MCLK	I2S master clock
GPIO43	I2S_BCK	I2S bit clock
GPIO44	I2S_DI	I2S data in (ADC → MCU)
GPIO45	I2S_WS	I2S word select / LRCK
GPIO46	I2S_DO	I2S data out (MCU → DAC)
GPIO47	PA_CTRL	NS4150B amplifier enable
GPIO48	FPC2 breakout	-
GPIO49	FPC2 breakout	-
GPIO50	FPC2 breakout	-
GPIO51	BAT_ADC	Battery/system voltage ADC (divider sample)
GPIO52–59	Reserved	Not assigned the IO;
GPIO60	SD-CTRL	SD 3.3 V power switch, active low

ECA/B　Reserved IO
Touch　LCD
SD　USB
Audio　FPC2 IO

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2.3 Display Information

Item	Specification	Unit	Remark
Display type	Round TFT, transmissive, normally black	—	Transmissive, Normally Black
Screen size	2.1	Inch	Diagonal
Resolution	360 × 360	Dots	—
Driver IC	GC9B72NA	—	—
Touch IC	CST826	—	Multi-touch, G+F CTP, AGC ≥6H, AF
Host interface	QSPI	—	As wired on core-board FPC1
Module OD (TP OD)	62.80 ± 0.1	mm	—
Viewing area VA	53.88 ± 0.2	mm	—
Active area AA	52.92	mm	—
Thickness	3.71 ± 0.2	mm	—
Brightness	300 (min) / 350 (typ)	cd/m²	—
Viewing angle	80/80/80/80 typ.	°	—
Logic supply VDD	3.3	V	—
Backlight	4× white LED	—	If≈80 mA, Vf≈2.8 V typ.
Operating temperature	−20 ~ 70	°C	—

2.4 Audio Interface

[Dock-dependent] This section lives on the dock. Core-only customers must implement an equivalent via FPC2 (I2S / AUDIO_I2C / PA_CTRL, etc.—see §2.5).

Dock-board audio solution: ES8311 codec + NS4150B amplifier + ZTS6216 analog microphone. Codec supply CODEC_3V3 is generated from 5 V by RT9013-33GB. Speaker is brought out on CN2 (1.25-2P WT).

Signal	GPIO / Device	Description
I2C SDA / SCL	GPIO40 / GPIO35	ES8311 I2C
I2S_MCLK	GPIO42	I2S master clock
I2S_BCK	GPIO43	Bit clock
I2S_DI	GPIO44	ADC → MCU
I2S_WS	GPIO45	LRCK / WS
I2S_DO	GPIO46	MCU → DAC
PA_EN	GPIO47	High = amplifier on

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MIC	ZTS6216	Differential ADC_MIC1P/N
Speaker	NS4150B	differential output, typ. 4 Ω / 3 W class

2.5 Board-to-Board Interconnect FPC2 (Customer Expansion Interface)

Use a 31-pin connecting cable with specifications of 0.3*31P*100mm. The core board expands the idle IOs for use by the carrier board (power, USB, audio, LED data, BOOT, BAT_ADC, EC2, etc.). Pinout matches the table below; The table lists signal pins already labeled on the schematic (consistent on core and dock).

Pin	Symbol	Description
1	GND	Ground (labeled on core side)
2	5V	Board-to-board 5 V supply
3	BOOTMODE	Download mode (→ GPIO61)
4	GND	Ground
5	GPIO49	To dock / candidate peripheral
6	GND	Ground
7	GPIO48	Candidate peripheral
8	GND	Ground
9	GPIO47	PA_CTRL
10	GPIO51	BAT_ADC
11	GPIO46	I2S_DO
12	GPIO50	Candidate peripheral
13	GPIO45	I2S_WS
14	GPIO39	Candidate peripheral
15	GPIO44	I2S_DI
16	GPIO38	Candidate peripheral
17	GPIO43	I2S_BCK
18	GPIO37	Candidate peripheral
19	GPIO42	I2S_MCLK
20	EC2_B	Second rotation-detection channel B (SSCM, core SW1)
21	GPIO40	AUDIO_I2C_SDA
22	EC2_A	Second rotation-detection channel A (SSCM, core SW1)
23	GPIO35	AUDIO_I2C_SCL
24	GND	Ground
25	USB_DN	USB differential −
26	USB_DP	USB differential +
27	GND	Ground
28	ESP_USB_P	Module USB +
29	ESP_USB_N	Module USB −

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30	5V	Board-to-board 5 V supply
31	GND	Ground
32	GND	Ground
33	GND	Ground

2.6 USB & Download Interface

[Dock-dependent] Type-C, ESD, CC resistors, and BOOT key are on the dock. Core board must implement power/download via FPC2 ESP_USB_*/USB_DP/DN, BOOTMODE, and 5 V/GND.

The Full kit has no CH340 or other USB-UART bridge; download and enumeration rely on ESP32-S31 native USB.

Port	Bridge / PHY	Role	Notes
Type-C	USB_DP/DN	Download / log / 5 V in	CC1/CC2=5.1 kΩ to GND (UFP); VBUS and differential lines have H5VL10B ESD; Hold BOOT (GPIO61), then tap RESET
	USB 2.0 HS PHY ESP_USB_P/N	OTG Host or Device	

Note: The schematic brings out both ESP_USB_P/N and USB_DP/DN differential pairs on Type-C (corresponding to plug A/B sides). software should treat them as module native USB. Recommended: hold BOOT → connect Type-C → use idf.py --preview flash monitor.

2.7 Rotation Detection / Keys / LED Ring

2.7.1 Overview

Rotary interaction on this product is not a classic EC11-class incremental quadrature encoder. Instead, the rotary mechanism drives two Alps Alpine SSCM-series compact two-way detector switches (SSCM110100) for direction/step sensing. Schematic net names still use EC1_A/B and EC2_A/B, which can be misread as incremental-encoder channels; this datasheet clarifies that here.

RefDes (core board)	P/N	Output nets	Role
SW2	SSCM110100	EC1_A / EC1_B	Primary detection path, direct to MCU GPIO
SW1	SSCM110100	EC2_A / EC2_B	Second detection path, via FPC2 to dock

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RefDes conflict note: Core-board SW1 = SSCM detector switch; dock-board SW1 = BOOT tactile key GT-TC018B-H035-L1. Same designator, different parts.

2.7.2 Device specification (SSCM110100)

Series: SSCM (Compact Two-way Detector Switch).

Item	Specification
Poles / positions	1 pole / 2 positions (two-way)
Outline size	4.0 × 5.0 × 1.5 mm
Total travel	1.57 mm
Operating force	0.35 N max
Rating (max)	1 mA / 5 V DC
Rating (min)	50 µA / 3 V DC
Contact resistance (initial)	2 Ω max
Life	50,000 cycles
Operating temperature	−10 ~ +60 °C
Circuit form	Common + contacts a / b (two directions)
Manufacturer note	Software must handle contact bounce / chattering

2.7.3 Keys & power switch

[Full kit / Dock-dependent] BOOT and SW3 are on the dock; Core-only must implement via FPC2.

Name	Location	P/N / function
BOOT (dock SW1)	Dock	GT-TC018B-H035-L1, pulls BOOTMODE (GPIO61) low; not the core-board SSCM SW1
Power slide SW3	Dock side	MST-12D18G4, controls VCC_OUT

2.7.4 LED ring

[Dock-dependent] LED ring is on the dock; Core-only must implement via FPC2 candidate GPIO / RGB_DATA.

Devices: 12×1010RGBC-2812B, daisy-chained.

Drive GPIO: RGB_DATA=GPIO61;RGB_DATA via R54 = 100 Ω; supply VDD_RGB / VCC_OUT.

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2.8 Storage & Battery Charging

Storage

Item	Description
On-board Flash	16 MB, 1.8 V domain
PSRAM	In-package 16 MB 1.8 V Octal SPI (ESP32-S31NRV16);
MicroSD	SDMMC 4-bit: CLK = GPIO24, CMD = GPIO25, D0–D3 = GPIO20–23

Battery & charging

[Full kit / Dock-dependent] Charge IC, battery connector, power path, and SW3 are on the dock. Core-only may use FPC2 5 V/GND/BAT_ADC for custom supply/fuel gauge.

Item	Description
Charge IC	TP4055 (U1)
Battery connector	CN1, 1.25-2P WT (BAT+)
Power path	slide switch SW3 selects/connects VCC_OUT
Voltage sense	BAT_ADC = 0.4 × VCC_OUT → GPIO51
nominal voltage	3.7V
Battery capacity	750mAh

2.9 Voltage & Current

Item	Conditions	Min	Typ	Max	Unit
Input voltage (USB)	Type-C VBUS	—	5.0	—	V
Core logic	TLV62569 output	—	3.3	—	V
Charge current	TP4055, Rprog=2.2 kΩ	—	≈470	—	mA
System operating current	Display on / Wi-Fi / audio, 5 V	—	600	—	mA
System standby current	Backlight off, RF idle	—	500	—	mA
Backlight LED current	Module spec 4 LED	—	≈80	—	mA

Recommended supply: Type-C 5 V / ≥1 A adapter (leave margin when charging and running the system together).

2.10 Reliability Test

Item	Conditions	Min	Typ	Max	Unit
Working Temperature	60%RH at 5V voltage	-20	25	70	C

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Storage Temperature	---	-30	25	85	C
Working Humidity	25°C	10%	60%	90%	RH
ESD	---	Contact: ±4KV Air: ±8KV			KV

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4.Functional Block Diagram

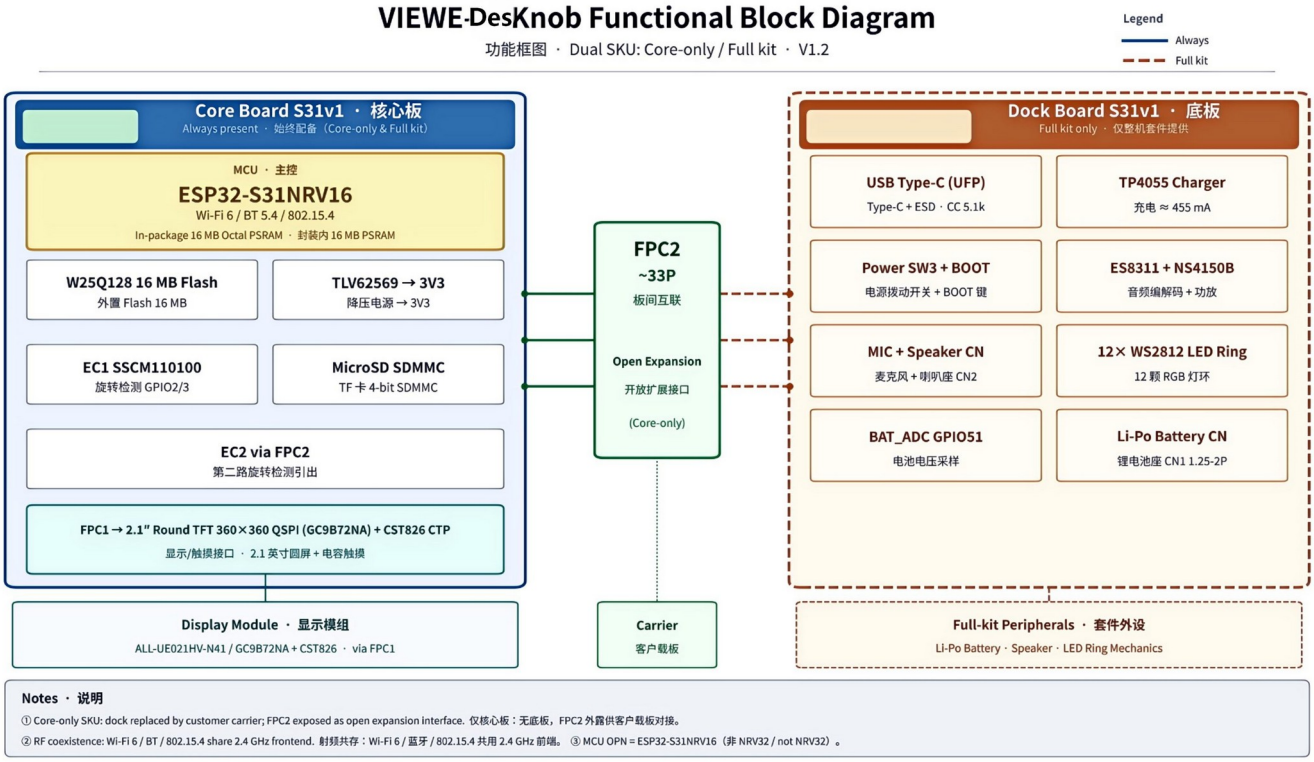


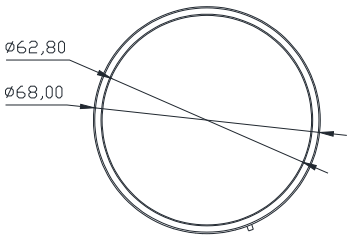
Figure 3-1 Functional block diagram

Note: ESP32-S31 includes a 2.4 GHz radio that supports Wi-Fi 6, Bluetooth 5.4 (LE), Bluetooth Classic, and 802.15.4. Because they share the same RF front-end, Wi-Fi and Bluetooth cannot transmit or receive simultaneously; the radio switches between protocols as needed. USB 2.0 High-Speed is an independent PHY and does not share that front-end.

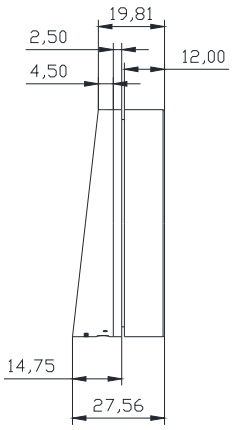
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4. MECHANICAL DRAWING

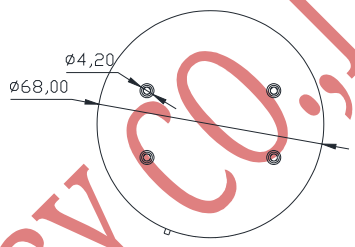
Front view

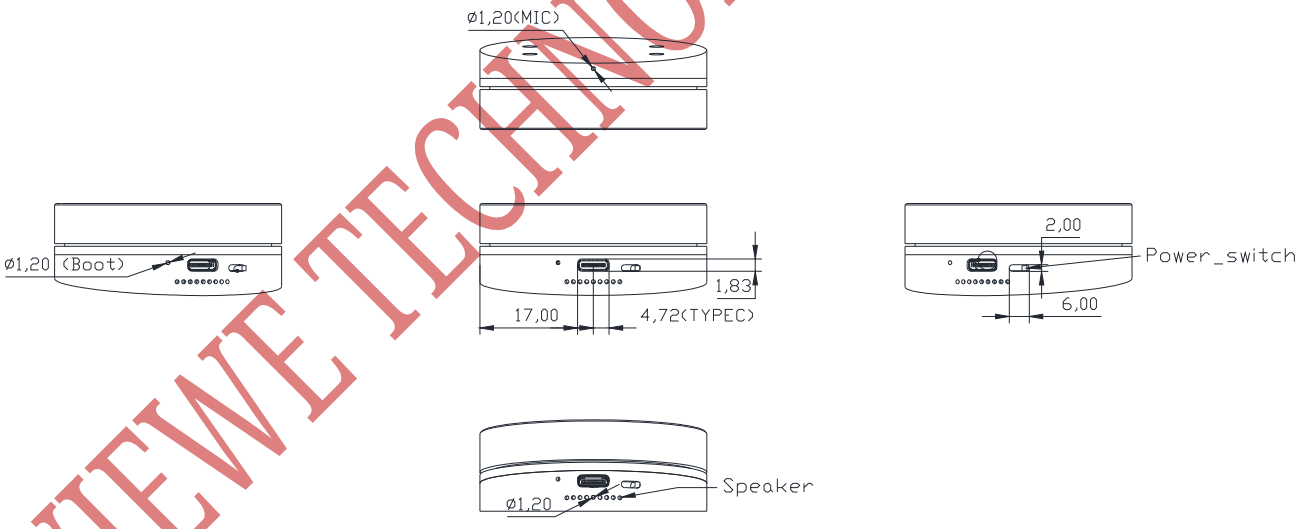


Side view



Back view





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5. Exploded structure



Figure 4-3 Mechanical exploded view

Stack illustration (rear to front): Rear housing → Li-ion Battery & Speaker → Dock Board → Light Guide Ring → Support Bracket → Rotating Bearing → Metal Housing → Core Board → 2.1" Round Touch Screen.

6. Detailed information (codes, drawings...)

Github：

Gitee:

Note: Relevant open-source materials and drawings are still being refined. Please stay tuned for subsequent versions. This document is based on facts extracted from schematic diagrams and the Espressif ESP32-S31 Preliminary Datasheet v0.5.

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